

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Differentiate between combinational and sequential circuit.
- Q.37 Explain the different addressing modes available in the 8085 instruction set. Write a simple assembly language program to add two 8 bit numbers stored in memory locations.
- Q.38 Minimize the following expression by using K-map and realize the result by using NAND gates.
- $$Y = \sum m(0, 1, 4, 5, 12, 13, 8, 9, 2, 6, 14)$$

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Q.1 What is the 2's complement of the binary number 1010?

a) 0101 b) 0110
c) 10111 d) 11010

Q.2 According to De Morgan's Theorem, A'B' is equal to

a) $\bar{A} + \bar{B}$ b) $A + B$
c) $\bar{A} \times \bar{B}$ d) $A + \bar{B}$

Q.3 A 4 bit shift register can store how many bits of data?

a) 1 bit b) 2 bits
c) 4 bits d) 8 bits

Q.4 Which of the following is a non-maskable interrupt in the 8085 microprocessors?

a) RST 7.5 b) RST 6.5
c) INTR d) TRAP

Q.5 Instruction MOV B,C is _____ byte instruction.

a) 0 b) 1
c) 2 d) 3

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- Q.6 A single transistor can be used to build ____ gates.
 a) OR gate b) NOT gate
 c) AND gate d) NAND gate
- Q.7 Which displays device consumes less power?
 a) LCD b) LED
 c) Fluorescent tube d) None
- Q.8 Algebra of logic is termed as
 a) Numerical logic b) Boolean algebra
 c) Arithmetic logic d) Boolean number
- Q.9 In the ASCII code system, how many bits are typically used to represent a character?
 a) 4 b) 7
 c) 12 d) 16
- Q.10 Which flip-Flop is commonly used to eliminate the race-around condition found in J-K flip-flops?
 a) R-S flip-flop
 b) Master slave J-K flip flop
 c) D-Type flip-flop
 d) T-Type flip-flop

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Prove that $A + A'B = A + B$
 Q.12 Why address and control buses are multiplexed?
 Q.13 A flip-flop is a memory element (T/F)

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- Q.14 Define T-State.
 Q.15 Draw symbol and truth table of OR gate.
 Q.16 Define dynamic RAM.
 Q.17 Define bus.
 Q.18 What is SISO?
 Q.19 Name the IC for DMA controller.
 Q.20 How many address lines are there in 8085?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What do you understand by EPROM and PROM? Explain.
 Q.22 Convert (127.25)₁₀ to octal number.
 Q.23 Simplify the expression $(A+B)(A+C)$
 Q.24 Write short note on ASCII code.
 Q.25 Draw and explain the circuit diagram and truth table for a full adder.
 Q.26 Explain the different addressing modes available in the 8085 instruction set.
 Q.27 Describe how multiple full adders can be cascaded to perform multi-bit binary addition.
 Q.28 Why address and control buses are multiplexed?
 Q.29 Explain with the help of diagram how you will convert a digital signal to analog signal.
 Q.30 Differentiate between latch and flip-flop.

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